

DEPI Graduation Project

Literature Review

System Under Test: Swag Labs (SauceDemo)

<https://www.saucedemo.com>

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Literature Review

This section provides the academic and industry context underpinning the testing methodologies applied to the Swag Labs project. It also defines evaluation criteria for the project.

2.1 Overview of Software Testing

Software testing is a systematic activity aimed at evaluating a software product to determine whether it meets specified requirements and to identify defects. According to ISTQB (International Software Testing Qualifications Board), testing encompasses verification and validation: verification ensures the product is built correctly, while validation ensures the right product is built.

For a web-based e-commerce application such as Swag Labs, testing is critical across functional and non-functional dimensions due to direct impact on end-user experience and transaction integrity.

2.2 Manual Testing Methodologies

Manual testing remains the foundation for exploratory testing and test case design. The following techniques are applied in this project:

- Equivalence Partitioning (EP): Dividing input domains into classes where all values are expected to behave equivalently (e.g., valid vs. invalid credentials).
- Boundary Value Analysis (BVA): Testing at the edges of input domains (e.g., maximum product quantity, minimum field lengths).
- Decision Table Testing: Mapping combinations of login conditions to expected outcomes.
- State Transition Testing: Modeling cart/checkout state changes (e.g., Empty Cart → Cart with Items → Checkout In Progress → Order Placed).
- Exploratory Testing: Unscripted investigation of application behavior, particularly for `locked_user`, `problem_user`, and `performance_glitch_user` roles.

2.3 Test Automation Frameworks

Selenium WebDriver is the industry-standard open-source tool for browser-based UI automation. When combined with TestNG and the Page Object Model (POM) pattern, it enables maintainable, scalable test suites.

- Selenium WebDriver: Provides API bindings (Java) to control browser actions, interact with web elements, and assert page states.
- TestNG: Provides test lifecycle annotations (`@BeforeClass`, `@AfterMethod`), parallel execution, data-driven testing via `@DataProvider`, and XML-based suite configuration.
- Page Object Model (POM): Encapsulates page-level UI interactions into dedicated classes, separating test logic from locator definitions — improving reusability and reducing maintenance overhead when UI changes.

- Maven: Dependency management and build tool integrating seamlessly with CI/CD pipelines.

2.4 API Testing Concepts

While Swag Labs is a front-end only demo application (no exposed REST API), best practices for API testing using REST Assured are documented as part of the academic scope. REST Assured is a Java DSL for simplifying HTTP request/response validation, supporting JSON Schema validation, authentication headers, and response time assertions.

2.5 Relevant Tools & Standards

Tool / Standard	Purpose
Selenium WebDriver 4.x	Browser automation
TestNG 7.x	Test execution framework
Maven	Build and dependency management
REST Assured	API test automation (academic scope)
JIRA / Excel	Bug tracking and test case management
GitHub	Version control and deliverable hosting
ISTQB Foundation Level	Testing methodology reference
IEEE 829	Standard for Software Test Documentation

2.6 Feedback & Evaluation

This section will be populated following the lecturer's formal review of the submitted project documentation. Feedback will be incorporated before the final submission deadline.

2.7 Suggested Improvements (Pre-submission Notes)

- Expand automation coverage to include negative test scenarios (e.g., injection attacks on login fields).
- Integrate test execution into a CI pipeline (e.g., GitHub Actions) to demonstrate DevOps alignment.
- Consider adding performance baseline measurements using browser DevTools or Lighthouse for NFR validation.

2.8 Final Grading Criteria

Category	Weight	Criteria
Documentation	25%	Completeness, clarity, adherence to DEPI guidelines, professional formatting
Requirements & Analysis	15%	Stakeholder analysis accuracy, use-case quality, FR/NFR completeness

Category	Weight	Criteria
System Design	15%	Diagram accuracy (UML, ERD, DFD), architecture rationale
Implementation	20%	Code quality, POM adherence, automation coverage, CI/CD usage
Testing & QA	15%	Test plan quality, bug report depth, coverage metrics
Presentation & Demo	10%	Clarity, slide quality, live demo, Q&A responses